

A-054

Roll No. 038

Total No. of Pages.: 03

21N502 /

B.Tech. II Sem Main/Back (New Scheme) Acad. Session 2023-24

All Branch

(2FY1-02) – Engineering Physics

21N502/

Time : 3 Hours

Maximum Marks: 70

Min. Passing Marks:

Instructions to Candidates:

Part – A: Short answer questions (up to 25 words) 10x2 marks = 20 marks. All 10 questions are compulsory.

Part – B: Analytical/Problem Solving questions 5x4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part – C: Descriptive/Analytical/Problem Solving Design questions 3x10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1 _____

2 _____

P.T.O.

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PART-A

- Q.1. What is thin-film Interference. [2]
- Q.2. In fraunhofer diffraction, what kind of source of light is used and where it is situated? [2]
- Q.3. During Population inversion, which process is dominant? [2]
- Q.4. What is the NA of the coaxial cable with its core and cladding indexes as 2.44 & 1.2? [2]
- Q.5. What is quantum mechanical tunneling? [2]
- Q.6. What is Wave function? [2]
- Q.7. Define energy bands in Solids. [2]
- Q.8. Write two applications of nano materials. [2]
- Q.9. In a wave if $E_0 = 100 \text{ V/m}$. Then find the magnitude of poynting vector. [2]
- Q.10. What is displacement current & its formula? [2]

PART-B

- Q.1. In a Newton's ring experiment, the diameter of the 20th dark ring was found to be 5.82 mm and the 10th ring is 3.36 mm. If the radius of the plano convex lens is 1m, calculate the Wavelength of light used. [4]
- Q.2. What are Einstein coefficients? Derive the expression for the relation between Einstein coefficients. [1+3=4]
- Q.3. For $n^2x + n^2y + n_3^2 = 26$ calculate the degeneracy, write the wave functions for the degenerated states and energy levels of degenerated states. [4]
- Q.4. A silver ribbon whose cross section is 1.0 cm by 0.2 cm. The ribbon carries a current of 100 A from left to right, and it lies in a uniform magnetic field of magnitude 1.5T. Using a density value $n = 5.9 \times 10^{28}$ electrons per cubic meter of silver, find the hall potential between the edges of the ribbon. [4]

- Q.5. A circular antenna of area 3m^2 is installed at a place in Madurai. The plane of the area of antenna is inclined at 47° with the direction of Earth's magnetic field. If the magnitude of Earth's field at the place is $40773.9 \times 10^{-9}\text{T}$.
Find the magnetic flux linked with the antenna. [4]
- Q.6. Derive the expression for time dependent Schrodinger wave equation. [4]
- Q.7. Draw the line diagram of Michelson's Interferometer and explain its construction. [4]

PART- C

- Q.1. Derive the expression of the intensity of light due to diffraction by a single slit. [10]
- Q.2. Describe the structure, types and features of optical fiber. [4+4+2]
- Q.3. Derive the expression of wave function and energy for the particle in a one dimensional potential box. [10]
- Q.4. Describe the properties of nano materials. Explain the synthesis of nanomaterials using top-down and bottom-up approach. [2+8]
- Q.5. Derive the maxwell equations in their integral form? [10]
